

Bitlocker-1

⚠ **Security Warning:** Work in an isolated environment (virtual machine or container). Never open unknown files on your primary machine.

Challenge Description

Jacky is not very knowledgeable about the best security passwords and used a simple password to encrypt their BitLocker drive. See if you can break through the encryption!

Hints

- Hash cracking
-

Tools Used

- `file` – Determine file type
 - `strings` – Extract printable character sequences
 - `exiftool` – Read metadata
 - `fdisk` – View partition tables
 - `dislocker` – Decrypt BitLocker-encrypted volumes
 - `bitlocker2john` – Extract BitLocker hashes for cracking
 - `john` (John the Ripper) – Crack password hashes
 - `mount` – Mount decrypted volumes
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Complete Solution

Step 1: Download and Prepare

Download the image named `bitlocker-1.dd` from the challenge page.

Step 2: Basic File Inspection

First, verify the file type and general information:

```
file bitlocker-1.dd
```

Expected output:

```
bitlocker-1.dd: DOS/MBR boot sector, code offset 0x58+2, OEM-ID "-FVE-FS-",
sectors/cluster 8, reserved sectors 0, Media descriptor 0xf8, sectors/track 63, heads
255, hidden sectors 124499968, FAT (32 bit), sectors/FAT 8160, serial number 0,
unlabeled; NTFS, sectors/track 63, physical drive 0x1fe0, $MFT start cluster 393217,
serial number 02020454d414e204f, checksum 0x41462020
```

The presence of `-FVE-FS-` (Full Volume Encryption File System) indicates this is a **BitLocker-encrypted** drive.

Step 3: Search for Visible Text

The flag may be present as plain text. Let's try extracting all character strings:

```
strings bitlocker-1.dd | grep -i "picoCTF"
```

Output: (no results)

⚠ No flag appears yet. The data is encrypted, so we need to decrypt the drive first.

Step 4: Inspect Metadata

Use `exiftool` to examine the file metadata for any hidden clues:

```
exiftool bitlocker-1.dd
```

Complete output:

```
ExifTool Version Number      : 13.50
File Name                    : bitlocker-1.dd
```

```
Directory          : .
File Size          : 105 MB
File Modification Date/Time : 2026:05:09 11:56:30+02:00
File Access Date/Time   : 2026:05:10 18:50:04+02:00
File Inode Change Date/Time : 2026:05:09 11:56:48+02:00
File Permissions      : -rw-rw-rw-
Error               : Unknown file type
```

The metadata itself is not suspicious.

Step 5: Examine the Partition Table

Let’s check the disk partition structure:

```
fdisk -l bitlocker-1.dd
```

Output:

```
Disk bitlocker-1.dd: 100 MiB, 104857600 bytes, 204800 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x78787878
```

Device	Boot	Start	End	Sectors	Size	Id	Type
bitlocker-1.dd1		2021161080	4042322159	2021161080	963.8G	78	unknown
bitlocker-1.dd2		2021161080	4042322159	2021161080	963.8G	78	unknown
bitlocker-1.dd3		4294932600	8589899894	4294967295	2T	78	unknown
bitlocker-1.dd4		4294967295	5035196669	740229375	353G	ff	BBT

Note: The partition type `78` is commonly used for **BitLocker encrypted partitions**.

Step 6: Attempt Decryption with Dislocker

First, create a mount point for dislocker:

```
mkdir /mnt/dislocker
```

Now try to decrypt the BitLocker volume:

```
dislocker -v bitlocker-1.dd /mnt/dislocker
```

Output:

```
Sun May 10 18:59:09 2026 [ERROR] No clear key found. Use a different method.  
Sun May 10 18:59:09 2026 [CRITICAL] None of the provided decryption mean is decrypting  
the keys. Abort.  
Sun May 10 18:59:09 2026 [CRITICAL] Unable to grab VMK or FVEK. Abort.
```

The disk image is protected by a **password**. We need to crack it.

Step 7: Extract BitLocker Hash with bitlocker2john

Use `bitlocker2john` to extract the hash information from the disk image:

```
bitlocker2john -i bitlocker-1.dd > bitlocker_hash.txt
```

Output:

```
Signature found at 0x3  
Version: 8  
Invalid version, looking for a signature with valid version...  
  
Signature found at 0x2195000  
Version: 2 (Windows 7 or later)  
  
VMK entry found at 0x21950c5  
....
```

The command extracts BitLocker hashes from the disk image and saves them to `bitlocker_hash.txt`. These hashes will be used to crack the password.

Step 8: Examine the Extracted Hashes

```
cat bitlocker_hash.txt
```

Output:

Encrypted device bitlocker-1.dd opened, size 100MB

Salt: 2b71884a0ef66f0b9de049a82a39d15b

RP Nonce: 00be8a46ead6da0106000000

RP MAC: a28f1a60db3e3fe4049a821c3aea5e4b

RP VMK:

a1957baea68cd29488c0f3f6efcd4689e43f8ba3120a33048b2ef2c9702e298e4c260743126ec8bd29bc6d58

UP Nonce: d04d9c58eed6da010a000000

UP MAC: 68156e51e53f0a01c076a32ba2b2999a

UP VMK:

fffce8530fbe5d84b4c19ac71f6c79375b87d40c2d871ed2b7b5559d71ba31b6779c6f41412fd6869442d66d

User Password hash:

\$bitlocker\$0\$16\$cb4809fe9628471a411f8380e0f668db\$1048576\$12\$d04d9c58eed6da010a000000\$60\$68156e51e53f0a01c076a32ba2b2999afffce8530fbe5d84b4c19ac71f6c79375b87d40c2d871ed2b7b5559d71ba31b6779c6f41412fd6869442d66d

Hash type: User Password with MAC verification (slower solution, no false positives)

\$bitlocker\$1\$16\$cb4809fe9628471a411f8380e0f668db\$1048576\$12\$d04d9c58eed6da010a000000\$60\$68156e51e53f0a01c076a32ba2b2999afffce8530fbe5d84b4c19ac71f6c79375b87d40c2d871ed2b7b5559d71ba31b6779c6f41412fd6869442d66d

Hash type: Recovery Password fast attack

\$bitlocker\$2\$16\$2b71884a0ef66f0b9de049a82a39d15b\$1048576\$12\$00be8a46ead6da0106000000\$60\$a28f1a60db3e3fe4049a821c3aea5e4ba1957baea68cd29488c0f3f6efcd4689e43f8ba3120a33048b2ef2c9702e298e4c260743126ec8bd29bc6d58

Hash type: Recovery Password with MAC verification (slower solution, no false positives)

\$bitlocker\$3\$16\$2b71884a0ef66f0b9de049a82a39d15b\$1048576\$12\$00be8a46ead6da0106000000\$60\$a28f1a60db3e3fe4049a821c3aea5e4ba1957baea68cd29488c0f3f6efcd4689e43f8ba3120a33048b2ef2c9702e298e4c260743126ec8bd29bc6d58

The most important hash for cracking is the **User Password hash**:

\$bitlocker\$0\$16\$cb4809fe9628471a411f8380e0f668db\$1048576\$12\$d04d9c58eed6da010a000000\$60\$68156e51e53f0a01c076a32ba2b2999afffce8530fbe5d84b4c19ac71f6c79375b87d40c2d871ed2b7b5559d71ba31b6779c6f41412fd6869442d66d

Note: We will use this hash because the recovery password hashes (48 characters) are much harder to brute-force.

Step 9: Crack the Password with John the Ripper

Create a new file containing only the user password hash and run John with the `rockyou.txt` wordlist:

```
john --wordlist=/usr/share/wordlists/rockyou.txt bitlocker.hash
```

Output:

Note: This format may emit false positives, so it will keep trying even after finding a possible candidate.

Using default input encoding: UTF-8

Loaded 1 password hash (BitLocker, BitLocker [SHA-256 AES 32/64])

Cost 1 (iteration count) is 1048576 for all loaded hashes

Will run 4 OpenMP threads

Press 'q' or Ctrl-C to abort, almost any other key for status

<REDACTED> (User Password hash)

Session aborted

John successfully cracks the password (the actual password is redacted here).

Step 10: Decrypt the BitLocker Volume

Use `dislocker` with the cracked password to decrypt the volume:

```
sudo dislocker bitlocker-1.dd -u<REDACTED> -- /mnt/dislocker
```

Note: Replace `<REDACTED>` with the actual cracked password.

Now check the mounted dislocker directory:

```
ls -l /mnt/dislocker
```

Output:

```
total 0
```

```
-rw-rw-rw- 1 root root 104857600 Jan  1  1970 dislocker-file
```

The `dislocker-file` is the decrypted NTFS filesystem image.

Step 11: Mount the Decrypted Filesystem

Create a mount point for the decrypted volume:

```
mkdir /mnt/bitlocker-1
```

Mount the decrypted file as a loop device:

```
mount -o loop /mnt/dislocker/dislocker-file /mnt/bitlocker-1
```

Output (expected warning):

```
The disk contains an unclean file system (0, 0).
Metadata kept in Windows cache, refused to mount.
Falling back to read-only mount because the NTFS partition is in an
unsafe state. Please resume and shutdown Windows fully (no hibernation
or fast restarting.)
Could not mount read-write, trying read-only
```

Note: The warning is normal. The disk was not properly shut down, so it mounts in read-only mode.

Step 12: Retrieve the Flag

List the contents of the decrypted volume:

```
ls /mnt/bitlocker-1
```

Output:

```
'$RECYCLE.BIN'  'System Volume Information'  flag.txt
```

Read the flag file:

```
cat /mnt/bitlocker-1/flag.txt
```

Final Result

picoCTF{<REDACTED>}

Note: The actual flag is redacted here. In the real challenge, the decrypted volume will contain the complete flag.

Key Concepts

- **BitLocker** | A full-disk encryption feature built into Windows.
- **FVE (Full Volume Encryption)**: BitLocker's encryption system, identified by the `-FVE-FS-` OEM-ID.
- **VMK / FVEK**: Volume Master Key and Full Volume Encryption Key – cryptographic keys used by BitLocker.
- **bitlocker2john**: A tool that extracts BitLocker hashes from encrypted disk images for password cracking.
- **dislocker**: A tool to decrypt BitLocker-encrypted volumes on Linux.
- **John the Ripper**: A password cracking tool that supports BitLocker hash formats.
- **Hash Cracking**: The process of guessing a password by comparing its hash to a known hash value.